

1 Miniaturisation~Small is in

Microsoft advocates the Tablet PC as the next small thing. Casio offers a number of computing functions in a wrist watch. Body implants can record a number of human functions. Growing popularity of the science of embedded chips using nanotechnology means that any device, however small it is, will be a computer at heart. New devices will contain integrated circuits a billion times smaller than today's chips, and yet be many times faster. The fusion of computers, networks and intelligent systems with nanotechnology will change the way business is done. It will give a whole new meaning to manufacturing. The knowledge involved in manipulating atoms and molecules with atomic level accuracy will define how manufacturers construct, distribute and duplicate substances known to man.

2 Unified Messaging~Speak my stuff

Natural Language Processing (NLP) is the stuff science fiction is made of—the ability of humans and machines to chat with each other like friends, with the machines displaying the ability to understand, analyse and generate articulate speech. While NLP is still a technology of the future, small measures are evident in the implementation of Voice over IP (VoIP) and speech recognition. VoIP enables conversation (or a video conference) using your instant messenger with another person on a traditional telephone line. Unified Messaging will consolidate VoIP, NLP and several other technologies to integrate different forms of communication (voice, fax, electronic transmission) and make them accessible from a variety of devices. For example, you can use a telephone system to call in to your office and log in to your e-mail account and have the messages read out using text to speech conversion. In a nutshell, unified messaging will lead to a reduction in costs as all forms of communication travel over data-networks.

3 Optical Fibre~Speeding things up

Fibre optic will change all notions of network speeds and make broadband a reality.

Fibre optic technology is already used in telephone infrastructure and undersea data links. The future will see fibre optics in the internal data paths of a computer mainboard enabling PCs to run faster and cooler. With widespread installation of fibre optic networks, the resulting surge in bandwidth availability will see a deviation in the way industries communicate. Video conferencing is one long-awaited tool that will emerge with a more extensive coverage of people. Once the proper infrastructure is in place, billions around the world will have access to this form of communication. Higher speeds have a direct correlation to time, and time saved is money saved.

4

Wireless and Mobile Technology~Instant access

802.11b, the next iteration of 802.11, promises 54 Mbps speeds (bettering the present standard by a factor of five) and addresses the security concerns in wireless transmission. 802.11b-based products are already out in the market and should see large-scale deployment by next year.

Portable devices such as cell phones and PDAs are just beginning to scratch the surface in mobile content delivery. The third-generation (3G) networks not only seek to unify the different mobile standards (GSM, GPRS, etc), but offer speeds ranging from 144 Kbps to 2 Mbps. Reaching out to those that are unwired enables businesses to widen their potential customer base. The wireless application platform has seen software developers queuing up to cash in on a multi-billion dollar industry.